

0 3 . 1

Explain how an arteriole can reduce the blood flow into capillaries.

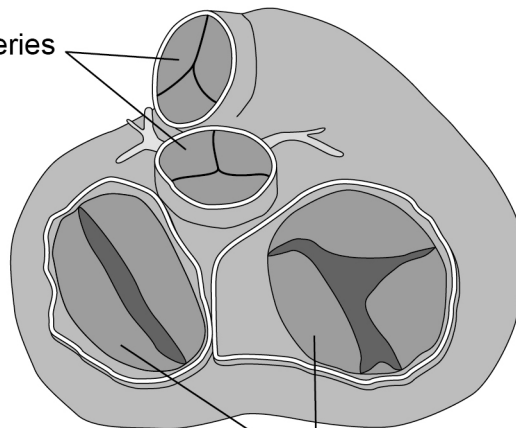
[2 marks]

Figure 1 shows heart valves during one stage of a cardiac cycle.

Ventricles are visible through the open valves.

Figure 1

Valves between
ventricles and arteries



Valves between
atria and ventricles



0 3 . 2

What can you conclude from the appearance of valves in **Figure 1** about heart muscle activity and blood movement between:

1. ventricles and arteries?

[2 marks]

2. atria and ventricles?

[2 marks]

Question 3 continues on the next page

Turn over ►



0 3 . 3

Tick (✓) **one** box next to the blood vessel carrying blood at the lowest blood pressure. [1 mark]

Capillary

☐

Pulmonary vein

☐

Renal vein

☐

Vena cava

☐

0 3 . 4

A scientist measured the heart rate and the volume of blood pumped in a single heart beat (stroke volume) of an athlete before exercise and calculated the cardiac output.

Cardiac output is calculated using this equation.

$$\text{cardiac output} = \text{heart rate} \times \text{stroke volume}$$

Her results are shown in **Table 1**.

Table 1

Heart rate / beats minute ⁻¹	Stroke volume / cm ³	Cardiac output / cm ³ minute ⁻¹
62	80	4960

After exercise, the athlete's stroke volume increased by 30% and the cardiac output was 13 832 cm³ minute⁻¹

Calculate the athlete's heart rate after exercise.

Give the answer to 2 significant figures. Show your working.

[2 marks]

Heart rate _____ beats minute⁻¹

